

Soft Optical Sensor for Bleeding Detection During Colonoscopy

Project Overview

Colonoscopy is an essential diagnostic and therapeutic procedure, but visualization is limited to the camera at the distal tip of the colonoscope. During navigation, the flexible colonoscope can loop or press laterally against the bowel wall, creating the possibility of bleeding outside the endoscopist's field of view. To address this limitation, our research developed a soft optical blood sensor that can be mounted directly onto a colonoscope and detect bleeding behind the distal-tip camera. The original platform used changes in optical transmission through a microfluidic sensing region to identify the presence of blood, while later work transitioned the technology into a wireless, flexible-PCB-based system.

The project evolved from a wired soft optical sensor into a more integrated wireless sensing platform. My contributions spanned sensor fabrication, ex vivo testing, fluidic-system development, clinician-facing software, PCB design and manufacturing, and exploratory localization using inertial sensing. Together, these efforts addressed different challenges involved in translating a benchtop sensing concept toward a system capable of functioning during realistic colonoscopy workflows.

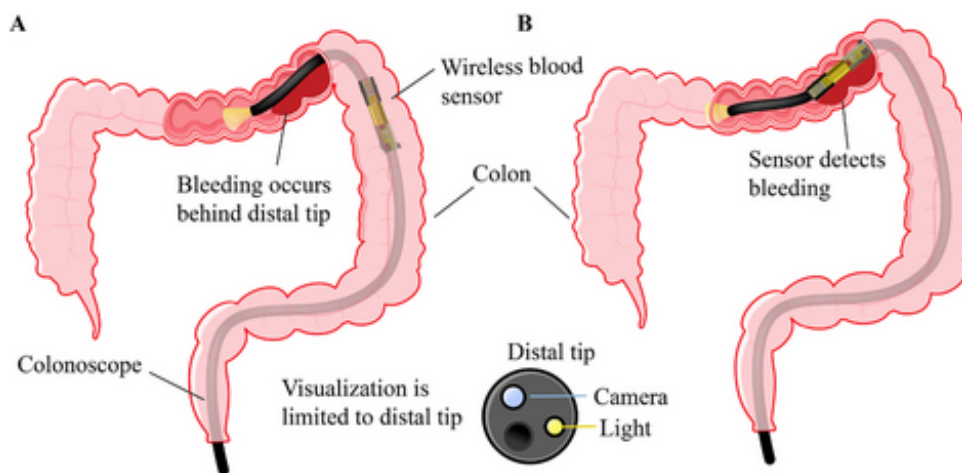


Figure 1. Clinical motivation and overall design of the soft optical blood sensor for colonoscopy. The sensor is mounted on the colonoscope to detect bleeding outside the distal-tip camera's field of view.

Initial Wired Sensor Development

My work on the project began with fabrication of the original wired soft optical blood sensor. The device used optical waveguides interfacing with a microfluidic channel so that fluid entering the sensing region altered the amount of transmitted light. The system used multiple wavelengths to distinguish blood from other changes in the optical signal, enabling real-time identification of bleeding while remaining

mechanically compatible with a flexible colonoscope. The original published system was specifically designed to detect blood at locations behind the distal-tip camera, where bleeding could otherwise remain unnoticed.

I contributed to the fabrication and assembly of these wired sensors, working with soft polymer components, optical interfaces, fluidic channels, and the electrical connections required for signal acquisition. This stage exposed me to the practical challenges of soft medical-device fabrication, including maintaining reliable optical alignment, preventing leakage, achieving repeatable bonding, and keeping the overall device sufficiently compact and compliant for integration onto a colonoscope.

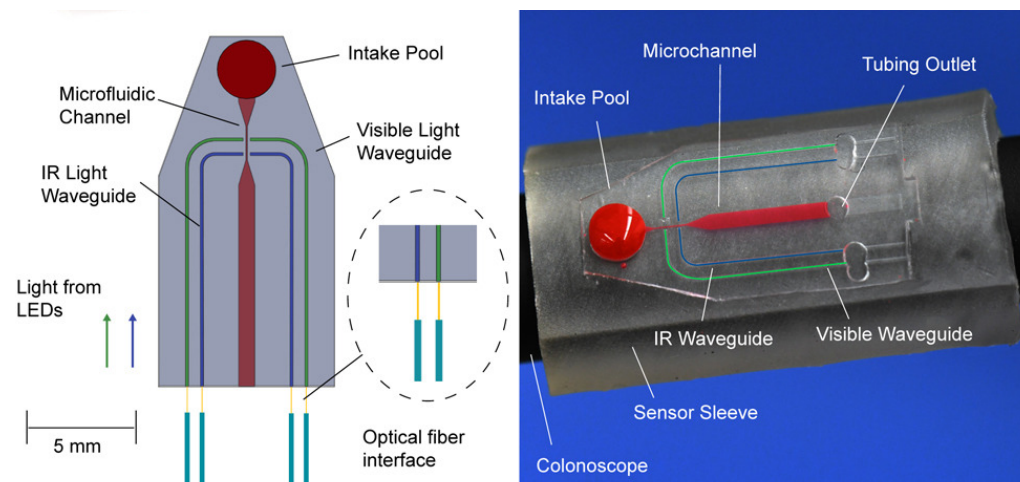


Figure 2. Architecture of the original wired soft optical blood sensor. The design integrates optical waveguides with a fluidic sensing channel to detect blood based on wavelength-dependent optical attenuation.

Ex Vivo Validation

A major stage of the project involved evaluating the sensor under realistic colonoscopy conditions rather than only on the benchtop. I participated in ex vivo experiments using a bovine colon model in collaboration with clinicians at Brigham and Women's Hospital. During testing, the sensor was mounted onto a colonoscope and navigated through the colon while bleeding events were introduced, allowing us to evaluate whether the device could detect blood during a realistic procedural workflow.

The 2024 study evaluated the soft optical sensor using an ex vivo bovine colon and a cohort of ten endoscopists, including both novice and expert users. The study showed that the sensor could rapidly identify bleeding while integrating into the colonoscopy workflow without significantly affecting navigation time or the users' learning curve.

For me, this stage was especially valuable because it demonstrated how a device that performs well on a laboratory bench can behave differently during actual endoscopic navigation. Issues such as sensor durability, tubing management, fluid contamination, bending, device attachment, and overall usability became important engineering constraints. Participating in these tests gave me experience evaluating a biomedical device in collaboration with clinicians and using experimental observations to guide subsequent design changes.

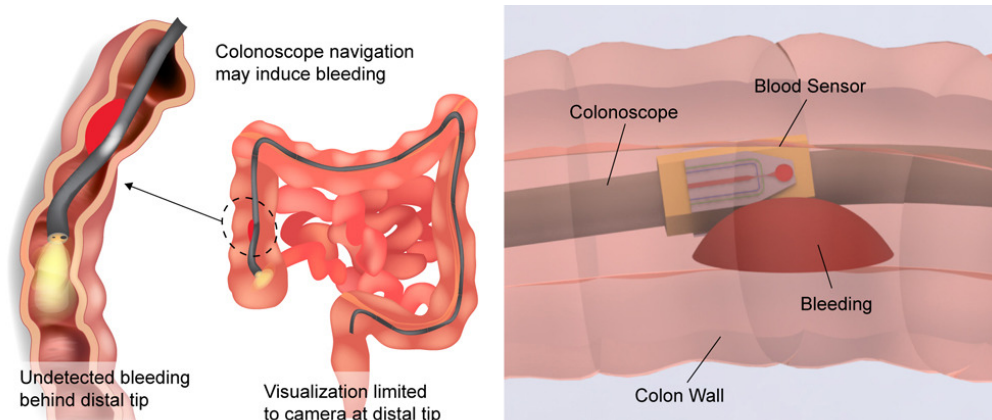


Figure 3. Ex vivo evaluation of the wired sensor in a bovine colon model. The sensor was mounted on a clinical colonoscope and tested under realistic navigation conditions to assess bleeding detection and device usability.

Fluidic System for Reliable Blood Sampling

One of the practical challenges identified during development was managing the biological fluids and residue entering the sensing region. Because the optical signal depended on fluid passing through the sensor channel, accumulated material could obstruct the optical path, interfere with blood detection, or compromise the sensor interface. I worked on developing the fluidic portion of the system to improve the reliability of sample transport through the sensor and reduce interruptions caused by accumulated fluid and residue.

This fluid-handling concept carried forward into the wireless generation. In the final wireless architecture, the milli-fluidic channel included an outlet connected to vacuum tubing, allowing fluid entering the sensor to be continuously suctioned away. The system also allowed air flushing when residue obstructed the fluidic passage. During ex vivo navigation, all three sensors were connected to pneumatic tubing to provide continuous suction and optional flushing.

My work on fluid management focused on making the sensor more robust in a wet and mechanically demanding environment. This included thinking about how fluids should enter and leave the sensing region, how suction or filtering could prevent accumulation, and how the fluidic interface could reduce the likelihood of sensing interruption or delamination. This work helped turn the device from an isolated optical sensor into a more complete optical-fluidic sensing system.

Real-Time Bleeding Detection Interface

Detecting blood optically is only useful clinically if the resulting signal can be translated into information that an endoscopist can understand quickly. I therefore worked on a graphical interface that processed incoming sensor measurements and indicated when a bleeding event was detected. Rather than requiring the clinician to interpret raw optical signals, the software provided a more intuitive representation of the sensor state.

The sensing principle relies on comparing optical responses at different wavelengths. In the wireless system, green light at approximately 520 nm is strongly attenuated by blood because of hemoglobin absorption, while infrared light at approximately 850 nm experiences much less attenuation. Comparing these responses allows the system to distinguish blood from other sources of signal change, including sensor deformation and gastrointestinal fluid. The processed sensor data can then be transmitted to an external receiver and displayed to the endoscopist.

Developing the interface gave me experience converting sensor measurements into clinically meaningful information, including signal processing, thresholding, data visualization, and user-interface design. It also reinforced the importance of designing not only the sensing hardware but also the way its output is communicated to the clinician.

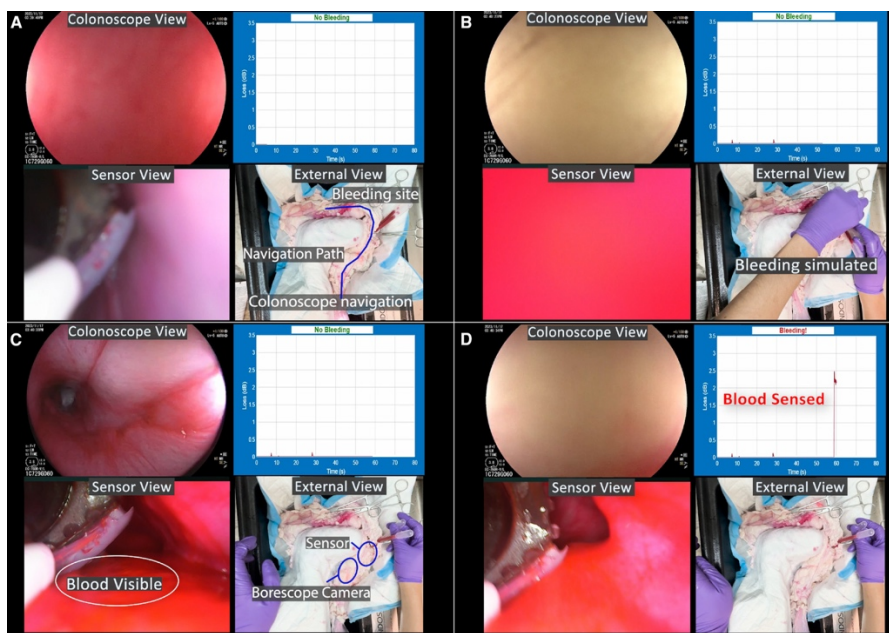


Figure 4. Custom graphical user interface for real-time bleeding detection. The interface converts sensor measurements into an immediate visual indication of whether bleeding is occurring.

Transition to a Wireless Sensor

As the project progressed, the design moved from a wired prototype toward a fully wireless sensing platform. Removing external wiring was important because cables running along the colonoscope could increase stiffness, interfere with navigation, and complicate integration into the clinical workflow. The wireless system instead incorporated surface-mounted electronic components, a microcontroller, wireless transmitter, and onboard batteries onto flexible printed circuit boards distributed across multiple soft sleeves mounted around the colonoscope.

During this transition, I learned PCB design and contributed to the design and optimization of the electronics used in the new sensor architecture. This required thinking beyond individual sensor components and considering the electronics as a complete system, including component placement, board

dimensions, connections to the optical sensing elements, power requirements, manufacturability, and the mechanical constraints associated with mounting the PCB on a flexible endoscopic device.

The final wireless architecture separated the optical sensing electronics, signal-processing and wireless circuitry, and power supply across three sleeves to limit the impact of the electronics on colonoscope flexibility. The sensor PCB incorporated LEDs and phototransistors, while additional circuitry supported signal acquisition, processing, and wireless transmission to an external receiver. This architecture allowed the system to operate without external wires running along the length of the colonoscope.

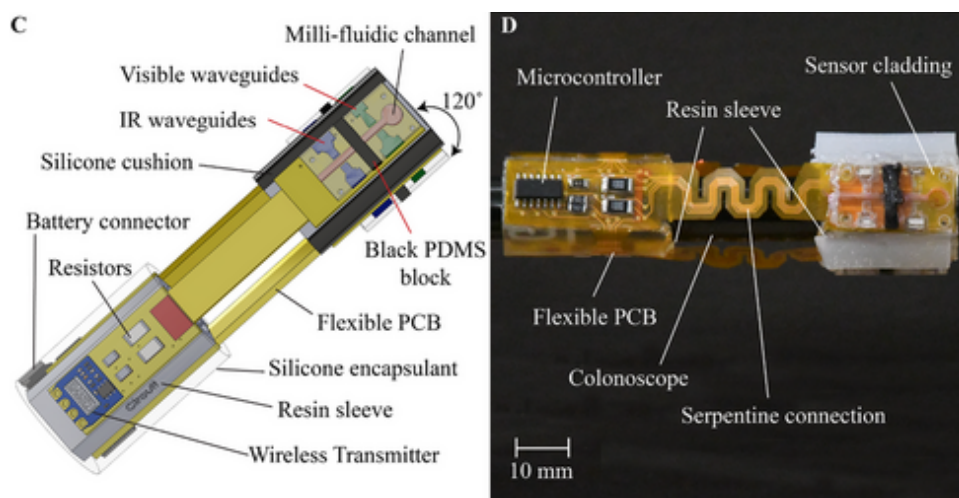


Figure 5. Wireless sensor architecture using flexible printed circuit boards. The updated system distributes sensing, signal-processing, wireless-transmission, and power components across flexible sleeves mounted on the colonoscope.

PCB Design and Manufacturing

Transitioning to the wireless platform required me to learn the complete workflow of printed circuit board development. I became familiar with schematic design, component selection, PCB layout, manufacturing constraints, and assembly considerations for a flexible medical-device platform. Unlike a conventional rigid PCB, the board needed to conform to the geometry of the colonoscope while keeping the sensor profile low enough to avoid significantly interfering with navigation.

The final device used surface-mounted components mounted directly onto a flexible PCB. The optoelectronic sensing board interfaced with the polymeric waveguides, while separate circuitry contained the wireless transmitter, microcontroller, and associated resistor networks. Components were assembled using solder paste, a stencil, and hot plate reflow before being integrated with the soft sensing structures.

This phase of the project was particularly valuable because it expanded my role from soft-material fabrication and signal processing into full electromechanical system design. I learned how layout decisions influence both electrical functionality and mechanical integration, and how to iterate a board design based on the practical constraints of a wearable or catheter-mounted system.

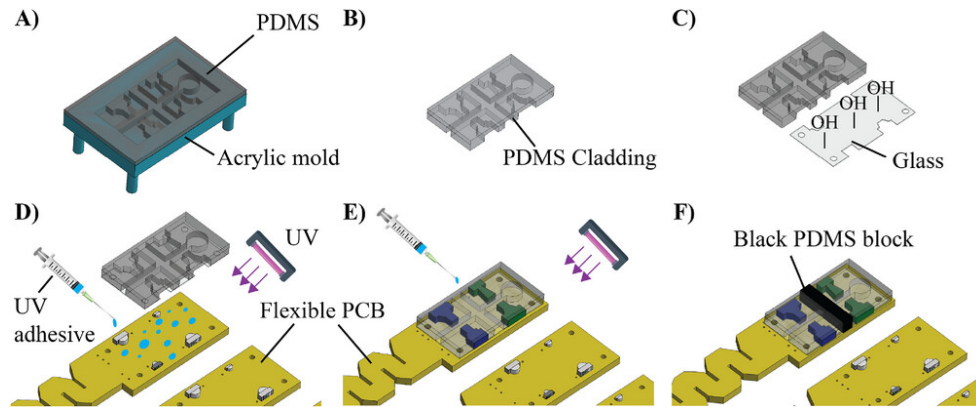


Figure 7. Fabrication and assembly workflow for the wireless soft optical blood sensor. The process includes PDMS molding, preparation of the sensor cladding, bonding to the flexible PCB, integration of the optical components, and final encapsulation of the assembled sensor.

Sensor Localization Using IMU Data

In parallel with the bleeding-detection system, I explored an additional research direction focused on estimating where bleeding occurred along the colon. Detecting blood alone informs the clinician that an adverse event may have occurred, but determining the approximate location of that event would make the system substantially more useful.

I developed an exploratory localization algorithm using inertial measurement unit data to estimate colonoscope movement and orientation. The approach incorporated accelerometer and gyroscope measurements and used a Kalman-filter-based sensor-fusion algorithm to reduce noise and improve estimates of the colonoscope's position and orientation. I also worked on visualizing the estimated path through a graphical interface so that sensor information could potentially be associated with a location along the colon.

This localization work was developed as a separate research direction and was not part of the published sensor results. However, the wireless publication identifies integration of an inertial measurement unit for more precise localization of bleeding as an important future direction for the platform, which aligns closely with the algorithmic work I explored.

Although IMU-based localization presented challenges including accumulated drift, sensor noise, and the difficulty of estimating translational position from inertial measurements alone, the project gave me significant experience with sensor fusion, Kalman filtering, motion tracking, and the limitations of real-world inertial navigation systems. It also helped motivate my later interest in data-driven and machine-learning approaches to biomedical sensing.

Final Outcome

The completed wireless system demonstrated that a soft optical sensor could be integrated onto a conventional colonoscope and provide real-time bleeding detection without significantly disrupting the

clinical workflow. The device combined optical sensing, milli-fluidic sampling, flexible electronics, wireless communication, and soft encapsulation into a compact add-on platform. In the published ex vivo study, the system successfully operated throughout 40 navigation trials and demonstrated 92.5% bleeding-detection accuracy, 95% sensitivity, and 90% specificity.

For me, this project was particularly valuable because my responsibilities evolved alongside the technology. I began by fabricating wired sensors and assisting with ex vivo validation, then worked on improving fluid handling and clinician-facing software, learned PCB design as the platform transitioned to wireless operation, and explored IMU-based localization as a potential extension of the sensing system. The project gave me experience spanning mechanical design, soft materials, electronics, signal processing, programming, experimental validation, and clinical collaboration.

Project Evolution

Over the course of the project, the technology progressed through several major stages:

Wired optical sensor → Ex vivo validation → Fluidic improvements → Real-time visualization → Flexible PCB design → Wireless sensing → Exploratory localization

My role evolved alongside the platform. I began with soft-sensor fabrication and ex vivo experimental testing, then worked on fluid management and real-time visualization, learned PCB design as the device transitioned to wireless electronics, and explored IMU-based localization as an additional research direction. This progression gave me experience across the full lifecycle of an experimental biomedical device, from fabrication and validation to electronics, software, system integration, and algorithm development.

Publications

Gerald, A., Palkawong Na Ayuddhaya, K., McCandless, M., Hsu, P., Pang, J., Mankad, A., Chu, A., Aihara, H., & Russo, S. “Ex vivo evaluation of a soft optical blood sensor for colonoscopy.” *Device*, 2(9), 100422, 2024. DOI: 10.1016/j.device.2024.100422. [PubMed](#)

Palkawong-Na-Ayuddhaya, K., Gerald, A., Chu, A., Pang, J., Mankad, A., Aihara, H., & Russo, S. “A Wireless Soft Optical Blood Sensor for Colonoscopy.” *Advanced Sensor Research*, 4(8), e00004, 2025. DOI: 10.1002/adsr.202500004.